adding horizontal growth to the old vertical scaling. EAIs were installed as a whole, as were the business applications installed within them. If the application was a monolith, the servers running it were even bigger monoliths. For example, applications such as Java Enterprise Application aRchives (EARs) that contained Web Application ARchives (WARs) and Java ARchives (JARs) had to be deployed entirely on a single server, such as WebLogic. These servers contained multiple services that had different hardware requirements. Some were very compute-intensive and needed high CPU power, while others needed mostly memory. Scaling was vertical—a larger machine for the single server. Thanks to containers, it was now possible to modularize services not logically, as usual, but relative to the infrastructure—and thus deploy them independently of each other.

In addition to all this containerization, the server side was decomposed. The various EAI implementations that ran the applications and orchestration were also broken into pieces. Project Borg, launched by Google in 2003, spawned many key ideas. Some of its developers led the way to Kubernetes, which was announced as an open source project in mid-2014 and became mainstream in just two years. With it, we have the routing, networking, scheduling, and security functions at the heart of any EAI. They not only orchestrate containers but also are built on top of them. Since then, instead of skyscrapers, software engineers have opted for a more down-to-earth architecture on all fronts. Scaling out is the order of the day.

## *B.1.3 Jailed services and other isolations*

With containers, you can increase the isolation of your components. This creates freedom in multiple dimensions, and it’s what revived the